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EXPERIMENT 15

Exp No:15 Experiment to understand the data preprocessing in Data science

AIM:
To understand the data preprocessing in Data science

DESCRIPTION:
Understand the importance of Data preprocessing in data science

CODE:

```
import numpy as np

import pandas as pd

# Read dataset

df = pd.read_csv("/content/Data.csv")

print(df)

# Display information about dataset

df.info()

# Find mode of Country

print(df.Country.mode())

# Get first mode value

print(df.Country.mode()[0])

# Type of Country mode

print(type(df.Country.mode()))

# Fill missing values

df.Country = df.Country.fillna(df.Country.mode()[0])

df.Age = df.Age.fillna(df.Age.median())
```

```
df.Salary = df.Salary.fillna(round(df.Salary.mean()))
print(df)
# One-hot encoding Country
print(pd.get_dummies(df.Country))
# Create updated dataset
updated_dataset = pd.concat(
    [pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
    axis=1
)
print(updated_dataset)
# Display dataframe information
df.info()
# Convert Purchased No/Yes to 0/1
updated_dataset.Purchased = updated_dataset.Purchased.replace( ['No', 'Yes'], [0, 1])
print(updated_dataset)
```

OUTPUT:

```
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0      NaN       Yes
5   France  35.0  58000.0       Yes
6    Spain   NaN  52000.0        No
7   France  48.0  79000.0       Yes
8  Germany  50.0  83000.0        No
9   France  37.0  67000.0       Yes
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
```

```
   Country   Age   Salary Purchased
0   France  44.0  72000.0        No
1    Spain  27.0  48000.0       Yes
2  Germany  30.0  54000.0        No
3    Spain  38.0  61000.0        No
4  Germany  40.0  63778.0       Yes
5   France  35.0  58000.0       Yes
6    Spain  38.0  52000.0        No
7   France  48.0  79000.0       Yes
8  Germany  50.0  83000.0        No
9   France  37.0  67000.0       Yes
```

```
0    France
Name: Country, dtype: object
France
```

```
   France  Germany  Spain
0    True    False  False
1   False    False   True
2   False     True  False
3   False    False   True
4   False     True  False
5    True    False  False
6   False    False   True
7    True    False  False
8   False     True  False
9    True    False  False
```

```
   France  Germany  Spain   Age   Salary Purchased
0    True    False  False  44.0  72000.0        No
1   False    False   True  27.0  48000.0       Yes
2   False     True  False  30.0  54000.0        No
3   False    False   True  38.0  61000.0        No
4   False     True  False  40.0  63778.0       Yes
5    True    False  False  35.0  58000.0       Yes
6   False    False   True  38.0  52000.0        No
7    True    False  False  48.0  79000.0       Yes
8   False     True  False  50.0  83000.0        No
9    True    False  False  37.0  67000.0       Yes
```

```
<class 'pandas.core.frame.DataFrame'>
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 0   Country    10 non-null     object
 1   Age        10 non-null     float64
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 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	False	True	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

RESULT:

Thus, data preprocessing was performed successfully using Pandas by handling missing values, filling them with appropriate statistical values, and applying one-hot encoding to categorical data. The Purchased column was also converted from Yes/No to 1/0 for further analysis.